

Benchmark Results and Evaluation Metrics

Final targeted FinReg benchmark: 160 hallucination stress-test questions covering false premises, unsupported source transfer, and unsupported operational detail.

Audited Benchmark Results

System Variant	Expected Behaviour	Point Coverage	Answer Rate	Abstain Rate	Forbidden Claim Hit	Mean Latency
Baseline RAG	80.00%	24.34%	100.00%	0.00%	0.00%	5.32s
RAG + Detector	92.50%	13.44%	46.25%	53.75%	0.00%	5.48s
RAG + Detector + Stochastic Gate	93.12%	13.28%	45.62%	54.37%	0.00%	5.88s

Benchmark Category Distribution

Question Category	Count	Evaluation Purpose
cross_source_nuanced	72	Reject unsupported transfer between regulators, documents, or contexts.
low_evidence_policy	40	Avoid inventing exact operational details from partial evidence.
false_premise	32	Reject misleading questions that assume fabricated requirements.
factual_supported	16	Answer normally when evidence directly supports the request.

Metric Definitions

Expected Behaviour Match	Main end-to-end quality metric: answer when supported, qualify/refuse when evidence is weak, and reject false premises.
Point Coverage	Secondary metric after audit because expected points are stricter atomic propositions, not short keyword labels.
Answer / Abstain Rate	Tracks usefulness versus safety blocking. A high answer rate can be risky on unsupported prompts.
Forbidden Claim Hit Rate	Direct hallucination-safety metric. Lower is better; 0.00% means no benchmark-defined forbidden claims were generated.
Mean Latency	Average runtime per benchmark question; captures the cost of detector and stochastic layers.

Key Finding

Baseline RAG answered every question and reached 80.00% expected behaviour, but still failed more often on unsupported-detail, source-transfer, and cautious-policy prompts. The detector raised expected behaviour to 92.50% by abstaining on high-risk answers. Stochastic uncertainty gating added a smaller positive gain to 93.12% while preserving 0.00% forbidden claim hits.